RTO and RPO

how long down, how much lost

Watch the 4-min video

In one line: RTO is the longest you may be down; RPO is the most data you may lose, measured in time. RPO drives how you copy data, RTO drives how you recover everything else, and lower numbers cost more, so set them per system from the business cost.

1 The two questions after every outage

How long were we down? How much data did we lose? Those answers have names. Agree them **before** the outage and they decide almost your whole recovery design (lessons 11–12) and its cost.

2 Definitions

RTO, recovery time objective: the maximum acceptable time from the start of an outage until service is restored. **RPO**, recovery point objective: the maximum acceptable data loss, measured in time. An RPO of one hour means you may lose up to the last hour of writes.

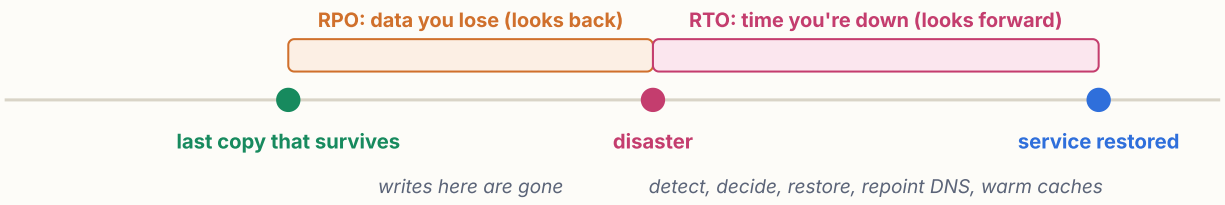


Figure 1. On one timeline, RPO looks backward from the disaster to the last copy that survives; RTO looks forward to recovery.

3 What each target forces you to build

Target	RPO needs (data)	RTO needs (everything else)
~24 hours	nightly backups	restore from backup, rebuild infra
~1 hour	hourly snapshots	pilot light: infra as code, data ready
minutes	continuous async replication or log shipping	warm standby, scripted failover
~zero	synchronous replication to another failure domain	active-active, automated failover

For RTO, count **every step**: detection, decision, restore or promotion, DNS changes, cache warm-up. A restore of a 2 TB database alone can take hours. RPO zero means each write is stored in two failure domains before it is acknowledged, which adds write latency (a cross-zone round trip on every commit).

4 Targets cost money, so set them per system

Synchronous replication slows every write; a hot standby doubles infrastructure. So targets belong to systems, not companies: payments might need **RPO 0, RTO minutes**; an internal reporting tool might accept **a day for both**. Tie each to the business cost of an hour down and an hour of lost data.

Then **test and monitor** them. Measure real restore times in drills, not estimates. Monitor **replication lag**: it is your real RPO right now. If lag reaches 10 minutes during a traffic spike, your 5-minute RPO is already broken; alert on it like an error rate.

WORKED EXAMPLE: AN RPO OF 5 MINUTES WITH NIGHTLY BACKUPS

The orders database has an RPO of **5 minutes** but only a 02:00 nightly backup. Lose it at 20:00 and up to **18 hours** of orders are gone. Fix: PostgreSQL continuous archiving, `archive_mode = on` and `archive_timeout = 60s`, ships the write-ahead log to object storage at least every minute, and a cross-region replica streams changes continuously. Worst-case loss is now about a minute (point-in-time recovery replays the WAL), and an alert fires if replica lag passes **3 minutes**.

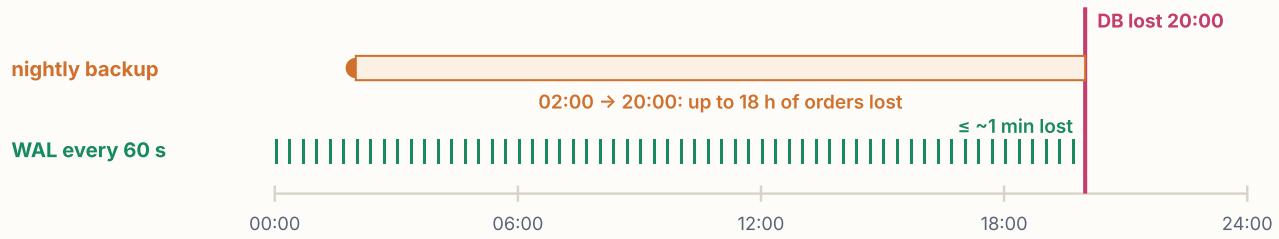


Figure 2. The same failure with two backup designs. The gap between the last surviving copy and the disaster is the data you lose.

COMMON MISTAKES

- Mixing them up: RTO is time down, RPO is data lost.
- One RTO/RPO for every system, usually too strict and too expensive.
- Promising an RPO while replication lag is not monitored.
- Counting only the restore, not detection, decision and DNS, in the RTO.

5 Key terms

RTO: maximum acceptable time to restore service

RPO: maximum acceptable data loss, measured in time

Synchronous replication: write acknowledged only once stored twice

WAL archiving: shipping the database's write-ahead log to storage

Point-in-time recovery: restore a backup, then replay logs to a moment

Replication lag: how far a replica is behind: your live RPO

6 Interview questions

Q1. What's the difference between RTO and RPO? BASIC

RTO is how long you can be down; RPO is how much data, measured in time, you can afford to lose.

Q2. What RPO do nightly backups give? BASIC

Up to 24 hours: anything written since the last backup is lost.

Q3. Your RPO is zero: what does the design need? SCENARIO

Synchronous replication to at least one other failure domain, so every acknowledged write already exists in two places, accepting extra write latency and cost.

Q4. The business wants RTO 15 minutes for a service restored from backups. Is it realistic? SCENARIO

Usually not: restoring and verifying a large database takes hours. You'd need a pilot light or warm standby with replicated data and scripted failover.

Q5. Your replica's lag hits 10 minutes during a spike. What does that mean for RPO? TRICKY

Your effective RPO is now 10 minutes; if the primary dies, that lag is lost. Alert on lag and fix capacity or throttling before it breaks the target.

Q6. Does RPO zero protect you from a bad DELETE? TRICKY

No: synchronous replication copies the mistake instantly. You need point-in-time recovery from backups and logs for that.

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